
Observability Debt in Agent Evaluation: Toward a Telemetry Layer for AgenticOS

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Abstract

Agentic AI systems are evaluated today through three disjoint public infrastructures—leaderboards, trace corpora, and telemetry logs—none of the public sources we examine jointly exposes the fields a resource-aware operating layer would need: task identity, model identity, scaffold and policy-configuration identity, outcome, token/cache usage, retries, latency, and billing. We measure its consequences across three studies on public data. First, using the Holistic Agent Leaderboard (HAL), raw dollar-cost rankings transfer across some benchmark pairs (Spearman $\rho=0.74\text{--}0.93$) but are mostly absorbed by a workload-persistent displayed-label cost propensity (pooled model $R^2=82.0\%$); success-adjusted cost is markedly less portable (mean $\rho=0.59$ vs. 0.84). Second, using 977 exact task–model–matched trajectory pairs from Open-SWE-Traces, scaffold choice changes observable execution burden at fixed task and model (OpenHands/SWE-agent turn ratio 0.85 , $[0.82, 0.87]$), and the widely assumed “expensive failure” premium is not detected under exact task matching (80%-power threshold: premium fractions ≥ 0.65 ; observed $0.44\text{--}0.49$). Third, auditing two independent telemetry corpora (8,058 and 1,017 sessions), we characterize the resource anatomy—93.7% mean cache-read share, $4.29\times$ conditional tool-error cascades replicated across sources, top-1%-of-sessions holding 48.3% of input-token volume—that a scheduler or router would need but that no benchmark currently attaches to an outcome. These studies are complementary rather than linked: Studies 1–2 show that transferred raw-cost rankings and unconditioned failure premiums confound *which workloads were attempted with how agents behaved on them*, while Study 3 shows that the execution variables hidden beneath those aggregates are themselves heterogeneous, heavy-tailed, and state-dependent. From this evidence we derive a joint telemetry schema spanning identity, execution, resources, control, and outcome, and argue it belongs at the operating-system layer: controlling execution and allocating resources requires observing what those allocations depend on.

1 Why This Belongs at the OS Layer

Agentic AI systems persist state, invoke tools, retry failed actions, and accumulate context across long horizons. Reasoning about their resource behavior—for scheduling, routing, cost budgeting, or governance—requires the same unified telemetry that operating systems have long provided for conventional workloads: a common schema linking *what ran*, *on what*, *under which policy*, *with what outcome*, and *at what resource cost*. We use *observability debt* to name the accumulated inability of public agent-evaluation infrastructure to jointly attribute resource consumption to task, model, execution policy, outcome, and system-level events. In every public source we examined, this debt prevents reliable attribution of observed resource differences to workload, model, scaffold, configuration, or execution behavior.

What makes this an *operating-system* responsibility rather than better benchmark bookkeeping? An agentic OS occupies a specific functional position. It observes *identity* (which task, model, scaffold, and policy configuration is running), *execution* (which tools act, what context accumulates), *resources* (tokens, cache, time, memory, money), *control* (retry, terminate, route, budget), and *outcome* (did the task succeed). The same resource-accounting principle motivates an OS-level treatment: the layer that controls execution and allocates resources needs access to the telemetry required to make those decisions. Without per-round token components, a memory manager cannot budget context growth; without configuration identity, a router cannot learn *why* one label is cheap; without error-status streams, a governor cannot halt a spiraling session. Today, each of these decisions is made implicitly from aggregate dollar figures (Figure 1).

We treat three public data sources as stand-ins for what this observability layer would unify: a cost-and-outcome leaderboard (HAL), a task/model/scaffold-matched trajectory corpus (Open-SWE-Traces), and raw session-level usage telemetry (two independent corpora). Each exposes a different partial view. HAL links dollar cost to outcome but encrypts execution traces. Open-SWE-Traces exposes exact task/model/scaffold matching and resolved status but no tokens or billing. The telemetry corpora expose token, cache, and tool-call dynamics but no outcome or billing. **No public source we examined jointly**

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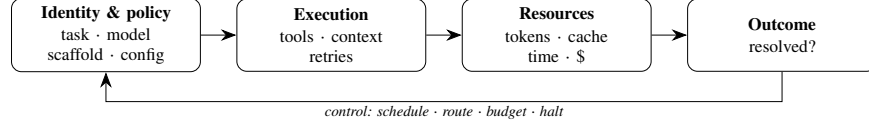


Figure 1: **The agentic OS control loop.** Each stage consumes exactly the schema fields derived in Section 6; the feedback arrow hosts scheduling, routing, budgeting, and halting decisions that today run on aggregate dollars because no examined public source closes this loop (Table 2).

44 **exposes task, model, scaffold and policy configuration, outcome, token/cache usage, retries, latency, and billing,** the
 45 joint schema a resource-aware OS layer needs (Table 2). Importantly, combining these sources does not create record-level
 46 observability: their different execution units, identifiers, and release boundaries prevent resource use, outcome, and billing
 47 from being joined into a single per-execution record.

48 Our three studies are therefore complementary rather than linked—each probes a different dimension of the gap, and none can
 49 be joined to another at the record level. (1) A coverage-aware cost-portability study (HAL) shows that aggregate dollar cost
 50 conflates displayed-label propensity with benchmark difficulty and decorrelates under success adjustment. (2) A matched-
 51 trajectory study shows that execution structure differs measurably across scaffolds, while a popular failure-cost heuristic turns
 52 out to measure workload composition. (3) A cross-source telemetry audit shows that the execution variables hidden beneath
 53 aggregate cost are themselves highly heterogeneous, heavy-tailed, and state-dependent. Together they triangulate one gap; no
 54 pair of them jointly identifies a causal mechanism. This paper is an empirical requirements study for an AgenticOS telemetry
 55 layer, rather than an implementation of the layer itself.

56 2 Related Work

57 Kapoor et al. (2026) introduce HAL as cost-aware evaluation infrastructure spanning coding, web, science, and customer-
 58 service tasks (21,730 rollouts, roughly \$40,000 in evaluation cost). Kapoor et al. (2024) motivate joint accuracy–cost report-
 59 ing and convex-hull frontier analysis, which we adopt as one of three decision-frontier definitions. Ndzomga (2026) asks
 60 whether task subsets preserve agent rankings within a single benchmark; we instead ask whether an *observed cost signal* pre-
 61 dicts a *different* workload. Benchmark² studies cross-benchmark rank consistency for static LLM benchmarks; our setting
 62 adds interactive scaffolds, dollar cost, success adjustment, and a held-out workload test. SWE-Effi reports that failed attempts
 63 consume substantially more resources than successful ones; our matched design tests that claim at fixed task and does not
 64 detect the premium once task-difficulty confounding is removed (Section 4). Open-SWE-Traces supplies the exact matching
 65 that makes this test possible but lacks token/billing fields, motivating Section 6. Commercial tracers expose per-deployment
 66 spans but are not cross-benchmark evaluation, and public releases strip execution detail, so the per-request visibility our
 67 audited sources lack is exactly what we quantify.

68 3 Study 1: Does a Cost Signal Travel Across Workloads?

69 3.1 Data and cohort

70 We use the public HAL CSV snapshot (Ndzomga, 2026): 242 displayed evaluations across nine benchmarks, 11 scaffolds,
 71 29 labels, supplying accuracy and aggregate dollar cost but no token components, retries, or per-task trajectories—the gap
 72 this paper is about.

73 The only broadly repeated scaffold is the displayed HAL Generalist Agent; fixing it and matching exact label strings yields
 74 six adequately overlapping workloads (CORE-Bench Hard, GAIA, SciCode, ScienceAgentBench, SWE-bench Verified Mini,
 75 TAU-bench Airline), a cohort of 86 rows and 25 labels.

76 3.2 Raw cost transfers; the mechanism behind it does not

77 Across six pairs with ≥ 12 shared labels, raw dollar-cost Spearman ρ ranges 0.74–0.93 (mean 0.84), surviving Holm-
 78 corrected permutation testing (Figure 2). A pooled label-plus-benchmark log-cost fixed-effects model explains $R^2=82.0\%$
 79 (adjusted 72.7%); once label and benchmark propensity are removed, **no high-overlap residual-cost association survives** a
 80 20,000-draw permutation test.

81 Leave-one-benchmark-out (LOBO) prediction succeeds for CORE ($\rho=0.78$, $N=14$, $p=0.0016$), GAIA (0.87, $N=16$),
 82 SWE-mini (0.90, $N=16$), TAU (0.83, $N=14$). It fails for SciCode (unstable, $[-0.50, 0.14]$) and ScienceAgentBench
 83 (directionally negative, $[-0.77, -0.43]$).

84 **Interpretation:** the portable component behaves as a workload-persistent displayed-label propensity—plausibly reflecting
 85 provider pricing, reasoning-mode defaults, or routing policy—that a router could in principle learn and reuse. but public data
 86 cannot identify *which*, since the distinguishing fields are not exposed.

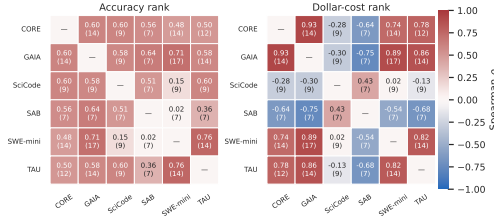


Figure 2: **Raw cost transfers; success-adjusted cost does not.** Pairwise Spearman ρ for benchmark pairs sharing ≥ 12 labels. Raw dollar cost is portable (mean 0.84); cost per expected success is not (mean 0.59), only three CPS pairs survive correction.

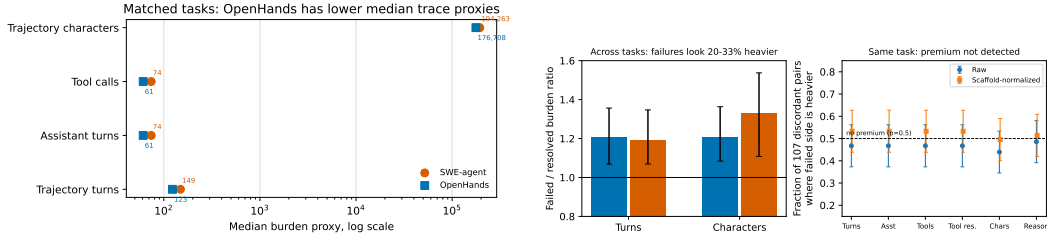


Figure 3: **(a) Scaffold burden on 977 matched pairs:** OpenHands uses $\sim 15\%$ fewer median turns/tool calls than SWE-agent at fixed task and model. **(b) The failure premium is not detected under exact task matching (80% power against $\geq 15pp$).** Left: naive cross-task ratios reproduce folklore (19–36% heavier failures). Right: among 107 discordant pairs, failed side heavier only 47–52/107.

3.3 Success-adjusted cost is a different, less portable ranking

Cost per expected success, $CPS_{i,b}(\epsilon) = C_{i,b} / \max(A_{i,b}, \epsilon)$ with primary floor $\epsilon=0.01$ (sensitivities at 0.001, 0.05, and zero-exclusion in the supplement), has mean $\rho=0.59$. Only three pairs survive correction: CORE–GAIA (0.87, [0.63, 0.96]), CORE–TAU (0.83, [0.31, 1.00]), GAIA–TAU (0.78, [0.42, 0.94]). The clearest raw-vs-CPS gap is GAIA–SWE-mini (0.60, [0.14, 1.13]): a label cheap in dollars is not reliably cheap per expected success elsewhere. No label tested on ≥ 5 workloads sits on every accuracy–cost frontier under discrete nondominance, convex hull, or 5% tolerance (o4-mini Low reaches 5/6 under tolerance; no universal winner under any strict definition).

Implication: Routing on displayed dollar cost therefore optimizes a signal whose portability degrades substantially once success is taken into account.

4 Study 2: What Matched Trajectories Reveal That Aggregates Cannot

HAL’s own trace archives are encrypted, so Study 1 cannot say *why* costs differ. We use an external unencrypted corpus—977 exact task–model pairs from Open-SWE-Traces for MiniMax-M2.5 under two scaffolds (OpenHands, SWE-agent), a reproducible but nonrandom four-shard boundary sample. The schema exposes resolved status, roles, tool calls, and text but not billed tokens or dollars; we treat turns, tool calls, and characters as trace-burden proxies.

4.1 Scaffold choice changes observable burden at fixed task

OpenHands has lower matched medians: 123 vs. 149 trajectory turns (ratio 0.85, [0.82, 0.87]), 61 vs. 74 tool calls (0.85, [0.82, 0.86]), 176,708 vs. 194,263 characters (0.95, [0.92, 0.98]). Resolved status shows no detected difference ($p=0.197$): scaffolds differ in execution structure without a detected accuracy trade-off here.

4.2 The “expensive failure” premium is not detected under exact task matching

Comparing failed vs. resolved trajectories *across* tasks reproduces SWE-Effi’s pattern (19–36% heavier failures). But among the **107 discordant pairs**—identical task and model, one scaffold resolved and the other failed—the failed side is heavier only 47–52 of 107 on every metric (binomial $p \geq 0.25$). An 80%-power pre-specified test against premiums $\geq 15pp$ finds the observed 0.44–0.49 uninformative for smaller effects; one model and a nonrandom sample bound the claim.

Interpretation: hard tasks make agents both burn more *and* fail more. An unconditioned failure multiplier measures *which tasks were attempted*, not a property of failing runs. Task-conditioned accounting requires exactly the joint schema no single public source provides.

Table 1: **Telemetry anatomy across two independent corpora.** Cache metrics are within-corpus; a share and a total-volume ratio are not interconvertible.

	TraceLab	MIMO Claude Code
Sessions / LLM rounds	8,058 / 665,453	1,017 / 4,690
Median input / output tokens per round	132,092 / 249	n/a (constant-tuple audit)
Cache-read vs. uncached input	93.7% mean share ($\approx 14.9\times$)	753.0M / 46.3M tokens = $16.3\times$
Median final-vs-initial context	$3.6\times$ (P90 $9.9\times$)	n/a
Top-1% / top-10% input-token share	48.3% / 88.2%	— / 37.9%
Error cascade $P(\text{err} \text{err})/P(\text{err} \text{succ})$	$21.0/4.9\% = 4.29\times$	4.3% vs. 1.0% ($4.33\times$)
Errored calls retried same-tool / retry succeeds	88.3% / 77.9%	—
Consecutive-error run length (median/P90/max)	1 / 3 / 368	—
Tool-call latency, success vs. error (median)	140 ms / 313 ms	—
Tool-result size, success vs. error (median chars)	602 / 208	—

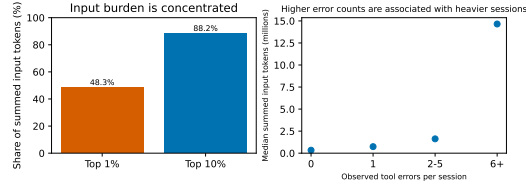


Figure 4: **Heavy-tailed burden.** Top-1%/top-10% summed-input concentration and median session input by observed-error bin (0.34M at zero to 14.65M at 6+).

5 Study 3: The Telemetry an OS Layer Would Need to See Directly

Studies 1–2 show aggregate cost hides *why* an agent is expensive and that execution structure varies across systems at fixed task and model. This section audits raw execution telemetry across two independent coding-agent corpora (Table 1). **TraceLab:** 8,058 sessions, 665,453 rounds, 743,819 tool calls. **MIMO Claude Code:** 1,017 sessions, 4,690 rounds. Neither links usage to outcome or billing.

Context is the dominant resource, not generation. The median TraceLab round carries 132,092 input tokens against 249 output tokens (93.7% mean cache-read share); median final-round context is $3.6\times$ initial (P90 $9.9\times$). A dollar total cannot separate price- from cache-policy- or accumulation-driven spend. Tool verbosity drives it: median next-input growth is 443 / 1,412 / 5,420 tokens below / within / above 10k tool-result characters.

Resource burden is heavy-tailed. The top 1% of sessions hold 48.3% of summed input volume; top 10% hold 88.2%. Median session input rises from 0.34M tokens at zero observed tool errors to 14.65M at six or more; mean-based budgeting would miss this tail-driven concentration (Figure 4).

Error cascades replicate across independent sources. After a successful call the next errs 4.9%, after an error 21.0% ($4.29\times$); MIMO replicates at $4.33\times$ (4.3% vs. 1.0%). Same-tool retry is common (88.3% get one, 77.9% succeed) but the tail is extreme (consecutive-error runs: median 1, P90 3, max 368). Failures are also slower yet less informative: 313 ms vs. 140 ms (P95 85.6 s vs. 15.8 s), returning 208 vs. 602 characters.

Data-quality boundary. MIMO repeats one constant per-round usage tuple, so its within-session token dynamics are not validated; we use it only for session counts, totals, concentration, and content-derived error sequences.

6 An Evidence-Derived Joint Telemetry Schema

Table 2 shows the coverage gap; the missing rows are exactly those Sections 3–5 show to be load-bearing. Each field is tied to an analysis above or to the OS control function it unblocks (*evidence-derived*: usefulness and necessity for those functions, not formal minimality). Keyed to that function:

1. (*Attribution / exact matching*) **Task and model identity** (exact IDs, not display strings)—enables exact-matched designs; without it the Section 4 test is infeasible on public data.
2. (*Routing between scaffolds*) **Scaffold/harness identity and version**—attributes burden differences to scaffold design rather than model behavior; input to routing.
3. (*Configuration attribution*) **Configuration and policy identity** (reasoning mode, sampling params, budgets, routing/cache policy)—distinct from model identity; Study 1’s unexplained propensity component is plausibly the shadow of these missing fields.
4. (*Success-conditioned accounting*) **Outcome/resolved status**—conditions any cost statistic on success, closing the CPS portability gap.
5. (*Memory / budget management*) **Token components per round (input, output, cache-read, cache-creation)**—separates price from accumulation and cache policy; lets a memory layer budget context growth.

Table 2: **The observability gap.** Coverage per public source examined: ■ exposed, □ partial (display-label-only, degraded fidelity, or inferable), – absent. No source covers the joint schema; the union leaves latency unlinked and round-level token fidelity unverifiable outside one corpus.

Field	HAL	Open-SWE	TraceLab	MIMO
Task identity	□	■	–	–
Model identity	□	■	□	■
Scaffold identity/version	■	■	□	■
Configuration / policy identity	□	–	–	–
Outcome / resolved status	■	■	–	–
Tokens / cache components	–	–	■	□
Retries / error status	–	□	■	□
Latency	–	–	■	–
Billing	■	–	–	–

- 146 6. (*Governance / halting*) **Tool-call counts, retries, error status per call**—detects and prices $4.29\times$ cascade dynamics; a
147 design motivation for governance policies that halt spiraling sessions—motivated here, not experimentally validated.
148 7. (*Real-time scheduling*) **Latency per call and per session**—present only as unlinked tool-call durations in one corpus;
149 required for real-time scheduling.
150 8. (*Procurement / auditing*) **Billing, linked to all of the above**—the one field HAL exposes well, and exactly the field that
151 is uninterpretable without the others.

152 Figure 1 places these fields at the stages of the control loop they close; Table 2 shows why, on public data alone, the loop
153 does not close.

154 No public source we examined exposes this joint schema. We do not propose a storage format or API—a systems-design
155 question outside our scope—but the measured consequences of its absence argue it belongs at a shared operating-system
156 layer rather than being re-derived piecemeal inside each benchmark, where today it is not derived at all.

157 7 Limits

158 This is a secondary-data analysis with clear boundaries. Display labels do not fully identify prompts, tools, budgets, or
159 harness versions; the HAL cohort is sparse, unbalanced, mostly single-run. Propensity adjustment is observational, not list-
160 price control; CPS is an aggregate ratio, not mean cost of successful runs. Study 2 uses one model and a nonrandom sample
161 whose proxies are not tokens or dollars, powered only against strong premiums ($\geq 15\text{pp}$). Study 3 is observational and
162 unlinked to outcomes or billing; MIMO’s per-round fields are unverifiable and used only where supported, and its design
163 implications are motivations, not evaluated policies. Each limit is itself an instance of the missing-field problem.

164 8 Conclusion

165 Across three studies, aggregate resource signals cannot be reliably attributed to workload, execution, or outcome without
166 telemetry fields the public sources we examined do not jointly expose. Raw-cost rankings transfer across workloads but are
167 strongly associated with a workload-persistent displayed-label cost propensity; success-adjusted cost reorders the ranking
168 (three of six pairs survive correction); the assumed failure-cost premium is not detected under exact task matching. No
169 examined source jointly links task, model, scaffold and configuration, outcome, token/cache usage, retries, latency, and
170 billing. The layer that controls execution and allocates resources is the natural owner of this telemetry, so we argue the
171 schema belongs at a shared operating-system layer for agentic AI rather than being reconstructed piecemeal inside each
172 leaderboard or trace release.

173 References

- 174 W. U. Ahmad, N. Ludwig, S. Majumdar, and B. Ginsburg. Open-
175 SWE-traces: Advancing dual-mode multilingual distillation for
176 software engineering agents. *arXiv preprint arXiv:2606.16038*,
177 2026.
178 Z. Fan, K. Vasilevski, D. Lin, et al. SWE-effi: Re-evaluating software
179 AI agent system effectiveness under resource constraints. *arXiv*
180 *preprint arXiv:2509.09853*, 2025.
- 181 S. Kapoor et al. AI agents that matter. *arXiv preprint*
182 *arXiv:2407.01502*, 2024.
183 S. Kapoor et al. Holistic agent leaderboard: The missing infrastruc-
184 ture for AI agent evaluation. *arXiv preprint arXiv:2510.11977*,
185 2026.
186 Franck Ndzomga. Efficient benchmarking of AI agents. *arXiv*
187 *preprint arXiv:2603.23749*, 2026.
188 Q. Qian, C. Huang, et al. Benchmark²: Systematic evaluation of
189 LLM benchmarks. *arXiv preprint arXiv:2601.03986*, 2026.

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